LifeOS Domain Systems: How the Logic Works

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Overview

This document explains how the systems of all 21 LifeOS domains actually work: what starts them, what data they read, the rules and steps they follow, and what they produce. The numbers given are design defaults you can tune.

Logic card. Each system is described in the same five parts:

- **Trigger:** what starts it (a chat message, a saved deadline, a daily job).
- **Reads:** the data it looks at.
- **Logic:** the rules or steps, in order.
- **Output:** what it proposes or records.
- **Guardrail:** what stops it from doing harm or acting alone.

Two principles hold for every system. First, the agent only proposes: every write goes through the calendar service, which checks conflicts and protected time and stores how to undo. Second, rules are code and wording is the model: exact rules (overlaps, minimum sleep, fixed times) run as tested functions, and the model only interprets your message and writes the explanation.

The shared engine

Every domain system runs the same eight steps. The domain only changes which data is read and which rules apply.

1. **Trigger.** A chat message, a saved deadline or exam, a manual calendar edit, or a scheduled job (morning brief, weekly review, daily check).
2. **Route.** The orchestrator picks the domain from the message or the event's domain, and loads that domain's prompt, config and top 5 memory notes.
3. **Read.** The system reads only its own domain tables plus the calendar, through `query_schedule`.
4. **Apply rules.** The domain's rules run as plain functions: conflicts, protected tiers, minimums, limits.
5. **Propose.** The system builds a draft: events, tasks, a report or a suggestion. Nothing is saved yet.

6. **Confirm.** A draft that touches protected time, or changes 5 or more events, is previewed and waits for you. Everything else is written straight away.
7. **Write and log.** The calendar service saves the change, and AgentLog stores the inverse so Undo restores the previous state.
8. **Measure.** Check-ins, hours and results flow into the domain's metrics, which feed the weekly review and the Reset reports.

Core rules reused across domains

Seven small rules do most of the work. Domain systems call them instead of reinventing them.

Rule	How it works	Used by
Backward planning	Takes a deadline and an hour estimate, splits the hours into blocks of 60 to 120 minutes, stops 1 day before the deadline for review, and places blocks in the best free slots	Education, Competitions, Finance, Admin, Blogging
Slot ranking	Removes any slot that overlaps a protected anchor, then scores the rest: higher for your focus hours, lower for an already heavy day, lower if it eats the buffer between blocks	Every system that places blocks
Spaced revision	Schedules reviews at growing gaps (1, 3, 7, 14 days) before a date, squeezes the gaps to fit short windows, and adds a review for any item rated weak	Education, Learning, Karate
Debt and repay	Debt is what fell short of a minimum over the last 7 days; the planner repays it by lightening upcoming days, never by adding load	Sleep, Rest and buffer
Load check	Sums planned hours per day and warns when the total passes your limit	Rest and buffer, all planners

Rule	How it works	Used by
Adherence	Done divided by planned for a period, counted only from check-ins and completed events	Habits, Reset reports, every domain's metrics
Drift detection	Fires when the same commitment is missed 3 days in a row, or adherence stays under 50% for 7 days	Habits, Reset Plan, Mood reflections

Study and work

Career

- **Application pipeline.** Each application is a card with a stage (saved, applied, screen, interview, offer, closed). Moving to a new stage creates the next task: a follow-up reminder 7 days after applying, prep blocks when an interview date is saved.
- **Contact log.** Each contact has a last-touch date and a follow-up date; the morning brief lists contacts whose date has passed.
- **Skill-gap radar (unique).** *Trigger:* you save a target role. *Reads:* skills named in the saved job posts, your Learning logs, your Portfolio evidence. *Logic:* count how often each skill appears across posts, mark each as covered (evidence exists), partial (learning logged) or missing, and rank missing skills by frequency. *Output:* a gap list and proposed learning blocks for the top 3. *Guardrail:* blocks are suggested until you accept them.

Education

- **Exam countdown.** A saved exam creates topics from the syllabus tree and runs backward planning with spaced revision, so topics get review sessions at growing gaps before the date.
- **Grade calculator.** Required score = (target grade − points earned so far) ÷ weight of the remaining work. The agent shows the number and never changes grades.
- **Mastery map (unique).** *Reads:* practice scores and self-ratings per topic. *Logic:* each topic gets a confidence from 0 to 100, raised after a good result and lowered as days pass without review; revision time is split in proportion to the gap from 100, weighted by exam weight. *Output:* a revision plan that leans on weak topics. *Guardrail:* you can override any topic's rating.

Competitions

- **Deadline tracker.** Each competition stores rules, eligibility and a submission time; the system flags overlaps between two competitions' build windows.
- **Submission checklist.** A template per competition type (deck, demo, repository, form) becomes tasks with due times that sit before the deadline.
- **Reverse sprint (unique).** *Trigger:* a submission time is saved. *Logic:* work backwards in order: submission, 2-hour buffer, demo recording, build freeze, pitch rehearsals, build blocks. Rehearsal slots avoid sleep and meals. The rubric scorer reads your pitch notes against the published criteria and lists the weakest criterion first. *Guardrail:* the scorer is advice on wording only, never a prediction of results.

Learning

- **Roadmap graph.** Each skill lists prerequisites; a node unlocks when all of them are marked done.
- **Resource library.** Resources have a status (to read, in progress, done) and a time estimate that sizes the learning block.
- **Skill tree (unique).** *Logic:* score each unlocked node = importance to your goals (from Career gaps and Education topics) + fit with free time this week − effort. The top node becomes the daily learning block. *Guardrail:* you can pin a node to override the score.

Open source

- **Issue and pull request tracker.** Each item has a status; a pull request with no update for 5 days creates a follow-up task.
- **Weekly contribution slot.** A recurring block placed in your focus hours.
- **Issue fit scorer (unique).** *Reads:* issue labels, size notes, your skill list, how fast a maintainer replied to your last contribution. *Logic:* fit = skill match + clear scope − estimated hours beyond your slot − slow maintainer response. *Output:* a ranked list and a reserved slot for the top issue. *Guardrail:* until GitHub sync exists, issues are entered by you.

Blogging

- **Pipeline board.** A post moves through idea, outline, draft, edit, publish; entering draft creates writing blocks sized by the outline.
- **Publishing checklist.** A fixed list (title, summary, links, proofread) that must be ticked before the post moves to publish.

- **Idea harvester (unique).** *Trigger:* the weekly review. *Reads:* the week's learning notes, post-mortems, merged pull requests. *Logic:* group notes by topic and propose a post for any topic with 3 or more notes. *Output:* idea cards with a suggested writing slot. *Guardrail:* nothing is added to the board until you accept it.

Portfolio

- **Evidence log.** Each entry stores what, when, a link, and which skills it proves.
- **Skills-to-evidence map.** For every skill, the count and recency of evidence; skills with none are shown as unproven.
- **Evidence auto-capture (unique).** *Trigger:* a competition result, a merged pull request, a volunteer milestone or a published post is recorded. *Logic:* turn it into a draft entry with skills inferred from its domain tags. *Output:* a proposed entry, and on request an application kit that selects the best evidence for a role. *Guardrail:* entries are private until you mark them public.

Body and health

Sleep

- **Sleep window and minimum.** You set a bedtime, a wake time and a minimum in hours. Any proposed change that leaves less than the minimum is flagged and needs your confirmation.
- **Focus hours.** Your chronotype setting marks the 3 to 4 hours you work best; slot ranking prefers them for demanding blocks.
- **Sleep debt bank (unique).** *Trigger:* a night is logged below the minimum. *Logic:* debt = sum of the shortfalls over the last 7 days. While debt is above zero, the planner removes the latest low-priority block from each of the next days and moves bedtime earlier by a small step, until the debt is repaid. *Output:* a repayment proposal. *Guardrail:* it never adds load to repay debt, and you approve the plan.

Food

- **Meal windows.** Protected blocks with a flexible start inside a window, so a meeting can shift a meal but never remove it.
- **Meal plan and grocery list.** The weekly plan lists meals; the grocery list is the sum of ingredients minus what you mark as in stock.
- **Prep-once planner (unique).** *Reads:* the week's load per day. *Logic:* label each day light, normal or heavy by planned hours; assign zero-cook meals to heavy days; place batch-cook blocks on the lightest day before them. In fasting mode, meal windows are

replaced by the windows Prayer supplies. *Guardrail:* no calorie or weight targets by default, and no nutrition or medical advice.

Karate

- **Syllabus by belt.** Each belt lists techniques and kata; each item has a status of new, learning or solid.
- **Conditioning and recovery.** A training week has a set number of hard sessions; the planner will not schedule a hard session the day after two in a row.
- **Belt path (unique).** *Trigger:* a grading date is saved. *Logic:* count the weeks left, divide the not-solid items across those weeks, give each week drill blocks for its items, and spend the last week on review only. Before a tournament it tapers: hard sessions are replaced by light ones. *Guardrail:* an injury note caps training load until you clear it.

Medicine

- **Medication schedule.** You enter each medicine, dose time and rule exactly as your clinician gave it. Dose times are fixed anchors, not flexible blocks.
- **Refill countdown.** Stock left ÷ doses per day = days remaining; a task appears when days remaining fall under your lead time.
- **Dose-safe scheduler (unique).** *Logic:* every other domain treats dose times as tier 1 anchors. If a change would clash with a dose, the system does not move the dose; it moves or asks about the other block. A reminder that goes unacknowledged is repeated once, then logged as missed with no judgment. *Output:* a visit brief that summarizes your own logs before an appointment. *Guardrail:* no diagnosis and no dosing advice; interaction warnings come only from a licensed drug database and say to ask a pharmacist or doctor.

Rest and buffer

- **Buffer rules.** After any block longer than a set length, a minimum gap is inserted; back-to-back blocks are flagged.
- **Load limit.** A daily cap on planned hours; passing it triggers a warning with 2 or 3 blocks to drop.
- **Recovery forecast (unique).** *Reads:* next week's planned hours, current sleep debt, 7-day mood trend. *Logic:* give each input a risk level of low, medium or high, and combine them so any two highs mean a high-risk week. *Output:* a suggestion to add buffer or lighten specific days before the week starts. *Guardrail:* a forecast is a prompt to rest, never a diagnosis.

Mind and spirit

Prayer

- **Tradition profile.** You choose your tradition, the calculation method it uses and which practices to track. The system holds no default faith and gives no rulings.
- **Time calculation.** Times come from your location, the date and the method you picked, using the sun's position, and are recalculated every day and when you travel.
- **Moving anchors (unique).** *Trigger:* each night, and whenever location or date changes. *Logic:* create the next day's prayer blocks as tier 1 anchors at the computed times, then re-run the conflict check on every other block that day; a clash is resolved by proposing to move the other block. Special days and fasting periods from your tradition's calendar notify Food (meal windows) and Sleep (earlier or split sleep). *Guardrail:* if the times cannot be computed, the system asks you rather than guessing.

Mood reflections

- **Check-in scale.** Mood and energy are each 1 to 5, optional notes, tags for context (people, place, activity). One tap takes about 10 seconds.
- **Pattern view.** Weekly averages and the days with the biggest swings; it shows your numbers and avoids labels.
- **Private weather map (unique).** *Reads:* your check-ins plus sleep hours, training days, planned hours and time outdoors. *Logic:* compare average mood on days with and without each factor; show a factor only with enough data points (design default: at least 10 days) and call it a pattern, never a cause. *Output:* a plain-language summary and an optional gentle mode that proposes lightening tomorrow. *Guardrail:* not therapy. Serious distress or any mention of self-harm pauses planning prompts and the agent responds with care and points to someone you trust or a local support line.

Habits and reflection

- **Habit list and check-in.** Each habit has a cadence and a daily done, skipped or missed result; Reset Plan commitments appear here without re-entry.
- **Streaks with grace.** A streak breaks only when skips in a week exceed the grace number you set.
- **Habit stacking.** For a new habit, the system finds an existing habit with a steady done rate (design default: 80% or more over 3 weeks) and proposes attaching the new habit to it, so it has a cue.

Forward planning

- **Direction and goals.** Yearly direction breaks into monthly goals; a goal must name an owner domain.
- **Life compass (unique).** *Trigger:* the weekly review. *Reads:* hours logged per domain and the intent you stated for each. *Logic:* gap per domain = intended hours − actual hours; list the 3 largest gaps and the domains that took the time instead. *Output:* a balance wheel and a proposed rebalancing for next week. *Guardrail:* intent is yours; the system never edits it.
- **Decision journal.** A decision stores your prediction and a review date; on that date the agent asks what happened and shows prediction next to outcome.

People and service

Volunteering

- **Shift scheduler.** A shift is a fixed event with travel and buffer blocks added by the Travel system.
- **Hours ledger.** Hours are summed per organization and month from completed shifts; a certificate task appears when a target total is reached.
- **Impact ledger (unique).** *Trigger:* a shift is marked done. *Reads:* the hours, the short impact note you add, and the role. *Logic:* roll notes into a summary by organization and skill, and compare planned volunteering hours with exam and competition weeks. *Output:* a shareable summary for CVs and applications, and a warning when hours crowd a heavy week. *Guardrail:* nothing is shared outside LifeOS unless you export it.

Social and mentors

- **Contact rhythm.** You set how often you want to be in touch with each person. Days since last contact > rhythm creates one soft reminder, then waits another full rhythm before asking again.
- **Mentor sessions.** Each session stores an agenda and notes; action items become tasks in the matching domain.
- **Mentor prep sheet (unique).** *Trigger:* a mentor session is within 2 days. *Reads:* open action items from the last session and progress in the domains you allow. *Logic:* list what was agreed, what moved, what is stuck, and suggest 3 questions. *Guardrail:* only domains you approve for this mentor are included, and sensitive domains are never included.

Life admin

Finance

- **Budget and expense log.** Each expense belongs to a category; a category's remaining amount = budget − spent, shown as you enter expenses. No investment, tax or other financial advice.
- **Savings goal.** Progress = saved ÷ target, with the pace needed per month to reach it by the date you set. The agent only reports the arithmetic.
- **Funding radar (unique).** *Trigger:* a scholarship, prize or stipend with a deadline is saved. *Logic:* run backward planning for essay, documents and review blocks, then mark each deadline as covered (blocks exist and fit) or at risk (not enough free time before it). *Output:* a monthly view of deadlines by status. *Guardrail:* it plans time, it does not judge your chances.

Admin

- **Renewal register.** Each item has an expiry or due date and a lead time; a task appears at the lead time.
- **Deadline ladder.** Reminders fire at 30, 14, 7 and 1 days before, and skip steps already passed.
- **Paper-trail builder (unique).** *Trigger:* a task such as "apply for a course". *Logic:* match the task to a checklist template, list each required document with its lead time (for example, time to get a letter), then schedule each one backward from the deadline, longest lead time first. *Output:* a checklist with dates. *Guardrail:* templates are starting points; you confirm the list because requirements differ by institution.

Travel and commute

- **Route times.** Each route stores a usual travel time; the system adds a buffer on top of it.
- **Time zones.** Events keep their own zone, and the calendar shows them in yours.
- **Door-to-door blocks (unique).** *Trigger:* an event with a location is saved. *Logic:* insert a travel block before it (route time + buffer) and one after it, then run the conflict check on those blocks too. *Output:* the calendar shows the real time cost. A packing list is generated from the trip type and the number of days. *Guardrail:* travel blocks are removed with the event they belong to.

Cross-domain systems

- **Protected tiers.** *Logic:* every block carries a tier. Tier 1 anchors (Medicine doses, Prayer times, Sleep minimum) can never be overlapped without your explicit confirmation. Tier 2 (meals, Karate, Rest) can move only inside a rule, such as a meal window. Tier 3 is flexible. When two blocks clash, the lower tier moves, and the system proposes the move instead of making it. You set the order of tiers 1 and 2.
- **Energy budget.** *Logic:* each day starts with a budget from last night's sleep against your minimum, the recent mood and energy check-ins, and the day before's load. Every block spends a cost by type (a demanding block costs more than a light one). When the day's total passes the budget, the agent proposes dropping or moving the lowest-priority block. The numbers are estimates that you can edit, and they never block a change you confirm.
- **Season profiles.** *Logic:* a profile is a saved set of rule overrides, for example exam season: shorter blocks, extra buffer, Karate reduced to light sessions. Turning one on applies the overrides to every domain at once and shows the list of what changed; turning it off restores the defaults. Profiles are switched by you, not by the agent.
- **Life graph.** *Logic:* every event, task, note and goal is a node linked to its domain, to the goal it serves and to related items. "Why is this block here" is answered by following the links, and clash checks use links to find items that depend on each other (a build block that serves a submission).
- **Evidence loop.** *Logic:* when a result is recorded in Competitions, Open source, Volunteering or Blogging, the loop creates a draft Portfolio entry and a Career note, and raises the matching skill in Learning's tree. Everything it creates is a draft you accept or discard.

Worked example: an exam week during a fasting period

You type: "Chemistry exam in 10 days, 12 hours of revision needed." You also have a fasting period starting in 4 days.

1. **Education** saves the exam, builds topics from the syllabus and runs backward planning: 12 hours become blocks of 90 minutes, finishing 1 day before the exam.
2. **Prayer** has already placed its moving anchors for each day, and **Sleep** holds its minimum. Both are tier 1, so slot ranking removes every slot that overlaps them.
3. **Food** sees the fasting period start and replaces the meal windows with the ones Prayer supplies; **Sleep** shifts bedtime and wake time inside your minimum.
4. **Season profile** "exam season" is offered; you accept, so Karate drops to light sessions and buffer grows.

5. **Mastery map** weights the blocks toward weak topics, and **spaced revision** adds short reviews at growing gaps.
6. **Energy budget** finds day 6 over budget because of a Karate class and 3 blocks; it proposes moving one block.
7. The agent shows the draft as dashed suggested blocks. You tap Accept all, and the calendar service saves it with an Undo button.
8. Each evening the **Habits** check-in records the blocks; if two days in a row go missed, **drift detection** suggests a lighter plan, and the next **Life compass** review shows how many hours Education actually received.

How the logic is tested

The rules are exact and the agent is not, so they are tested differently.

Layer	What is tested	How
Rule functions	Backward planning, slot ranking, debt, load, adherence, drift, time calculation	Unit tests with edge cases: midnight, back-to-back blocks, daylight saving, time zone changes, empty data
Domain systems	Each logic card's steps end to end, such as Belt path from a grading date to weekly blocks	Fixture data per domain and a check of the resulting calendar
Agent behavior	The right tool is called with the right arguments	30 to 50 saved commands per wave, replayed and compared with expected calls
Safety	Tier 1 anchors never overlapped without confirmation, no medical advice, distress handling, sensitive domains excluded from other reports	Dedicated cases that must always ask or refuse
Undo	Every write can be reversed	One undo test per tool and per system that writes